

ASSIGNMENT #9 November 1, 2022

COORDINATOR: STEFANO FOCESATTO

7.1.11, 7.1.7, 7.1.14

SECTION 7.1

Exercise (7.1.7 — Stefano Fochesatto). The center of a ring R is $\{z \in R \mid zr = rz \text{ for all } r \in R\}$. Prove that the center of a ring is a subgroup that contains the identity. Prove that the center of a division ring is a field.

Proof. Suppose a ring R with identity and let $G = \{z \in R \mid zr = rz \text{ for all } r \in R\}$. Note that by definition we get that $1 \in G$ since $1r = r1$ for all $r \in R$. Let $a, b^{-1} \in G$ and $r \in R$, note that $(a + b^{-1})r = (ar + b^{-1}r) = (ra + rb^{-1}) = r(a + b^{-1})$. Therefore $G \leq R$ with respect to addition. Note that for all $r \in R$ we have that $(ab)r = (ar)b = r(ab)$. Therefore G is closed with respect to multiplication and thus is a subring of R .

Suppose R is a division ring and let $g \in G$ and note that $g^{-1}r = (r^{-1}g)^{-1} = (gr^{-1})^{-1} = rg^{-1}$ so $g^{-1} \in G$. Therefore G is a commutative division ring, and therefore a field. $\square$

Exercise (7.1.11 — Stefano Fochesatto). Prove that if R is an integral domain and $x^2 = 1$ for some $x \in R$ then $x = \pm 1$.

Proof. Suppose R is an integral domain with $x^2 = 1$ for some $x \in R$. Note that $x^2 - 1 = 0$, so therefore $(x - 1)(x + 1) = x^2 - 1 = 0$. By definition R has no zero divisors, therefore $x - 1 = 0$ or $x + 1 = 0$ so $x = \pm 1$. $\square$

Exercise (7.1.14 — Stefano Fochesatto). Let x be a nilpotent element of the commutative ring R .

(a) Prove that x is either zero or a zero divisor.

Proof. Suppose $x \in R$ is nilpotent and R is commutative. Let m be minimal where $x^m = 0$ then it follows that $xx^{m-1} = 0$. Suppose $x \neq 0$, since m is minimal $x^{m-1} \neq 0$ and therefore x is a zero divisor. $\square$

(b) Prove that rx is nilpotent for all $r \in R$.

Proof. Suppose $x \in R$ is nilpotent and R is commutative. Let m such that $x^m = 0$. Note that $(rx)^m = r^m x^m = r^m(0) = 0$ thus rx is nilpotent. $\square$

(c) Prove that $1 + x$ is a unit in R .

Proof. Suppose $x \in R$ is nilpotent and R is commutative. $(1+x)(1+\sum_{i=1}^{m-1}(-1)^i x^i) = (1)(1+\sum_{i=1}^{m-1}(-1)^i x^i) + (x)(1+\sum_{i=1}^{m-1}(-1)^i x^i) = 1 + \sum_{i=1}^{m-1}(-1)^i x^i + \sum_{i=1}^m(-1)^{i+1} x^i = 1 \pm x^m = 1$. $\square$

(d) Deduce that the sum of a nilpotent element and a unit is a unit.

Proof. Suppose $x \in R$ is nilpotent, $r \in R$ is unit and R is commutative. There exists an $a \in R$ such that $ra = ar = 1$, and some m such that $x^m = 0$. By (b) we know that ax is also nilpotent. By (c) we know that $(1 + ax)$ is a unit such that for $b \in R$, $(1 + ax)b = b(1 + ax) = 1$. Finally note that $(x + u) = (xa + 1)u$. Multiplying by (ab) , since R is commutative we get, $(x + u)(ab) = (xa + 1)u(ab) = ((xa + 1)b)(ua) = 1$ $\square$